

Mingrui Li

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Biostatistics MPH candidate with hands-on experience analyzing **clinical trial data**, **healthcare claims data**, and **high-dimensional genomic data**. Builds **reproducible analytical pipelines (R, SAS, SQL)** and translates statistical findings into **interactive tools and clear reports** that support clinical research and operational decision-making, with particular depth in **predictive modeling**, **survival analysis**, and **healthcare cost and utilization analytics**.

EDUCATION

Emory University

Master of Public Health, Biostatistics

Aug. 2024 – Expected May 2026

Atlanta, GA

University of Liverpool

Bachelor of Science, Bioinformatics

Sep. 2020 – May 2024

Liverpool, UK

EXPERIENCE

Emory University

Research Data Analyst — Translational Genomics

Jan. 2025 – Jan. 2026

Atlanta, GA

- Analyzed gene expression data across 54 human tissues (n=981 samples), building a statistical pipeline with multivariable regression, confounder adjustment, and FDR correction to identify tissue-specific aging biomarker candidates.
- Productized findings into the *Age Effect Explorer*, a SQL-backed Shiny application that reduced query time from >3 minutes to <10 seconds, enabling real-time hypothesis exploration for 10+ biology collaborators without writing code.
- Co-first authored a peer-reviewed publication; managed project from conception through analysis, cross-functional coordination, and manuscript preparation.

SELECTED PROJECTS

Medicare Provider Payment Variation Analysis

Sep. 2025 – Mar. 2026

Emory University

Atlanta, GA

- Analyzed national Medicare claims data spanning 3,000+ hospitals and 600+ clinical procedures to quantify DRG-level cost and utilization variation across providers.
- Applied statistical outlier detection (IQR- and z-score-based) to surface anomalous payment and utilization patterns from high-dimensional claims data.
- Built an interactive Tableau dashboard translating complex variation patterns into accessible visualizations for clinical and operational stakeholders.
- Optimized BigQuery SQL queries for scalable processing of large-scale claims records while maintaining analytical rigor.

Clinical Trial Survival Analysis & EEG Biomarker Evaluation

Mar. 2025 – Dec. 2025

Emory University

Atlanta, GA

- Executed survival analysis on multi-center clinical trial data, quantifying treatment efficacy in delaying seizure onset through Cox regression, hazard ratios, and Kaplan-Meier curves.
- Collaborated with clinical investigators to define event and censoring rules, constructing analysis-ready datasets from raw trial data for time-to-event modeling.
- Built an EEG-based biomarker classifier to stratify seizure risk; evaluated performance using sensitivity, specificity, and AUC to support clinical decision-making.
- Delivered reproducible SAS pipelines and clinical reports bridging statistical findings with actionable recommendations for investigators.

Breast Cancer Biomarker Classification Pipeline

Jan. 2026 – May 2026

Emory University

Atlanta, GA

- Built an ML classification pipeline in R to predict tumor regions from spatial gene expression data across 94 breast cancer tissue samples (~9,000 spots × 14,019 features), achieving AUC = 0.89 via grouped 5-fold cross-validation.
- Implemented memory-efficient processing with sparse matrix storage and subsampling to handle high-dimensional genomic data within compute constraints.
- Applied SHAP explainability to identify key predictive biomarkers; validated findings through genome-wide statistical testing and communicated results via volcano plots and GO enrichment visualizations.
- Packaged the full analysis in a reproducible pipeline (Docker, renv, Makefile) with integrated R Markdown reports documenting methods, findings, and limitations.

TECHNICAL SKILLS

Programming & Data: R (tidyverse, Shiny, survival, caret), SAS, SQL (BigQuery), Python (pandas, scikit-learn), Linux/Bash

Analytics & Methods: Regression, survival analysis, predictive modeling (cross-validation, SHAP), cohort analysis, PCA, diagnostic accuracy, statistical outlier detection

Visualization & Reporting: Tableau, R Markdown, Jupyter Notebook, Microsoft Excel/PowerPoint

Reproducibility & Infrastructure: Docker, renv, Makefile, Git